

ReMark: Enhancing Deepfake Forensic Watermark via Image Reconstruction on Latent Space

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Abstract—Deep invisible digital watermark is widely used in media forensics, yet recent deepfake techniques can severely distort the embedded watermark signal. Prior work largely strengthens watermarking frameworks, but even the most robust “Vase” can be shattered by repeated “Hammer” attacks. We take a different perspective: rather than continually designing stronger embedding schemes, we repair the watermark itself. Specifically, we focus on GAN-based deepfake attacks and propose a plug-and-play restoration method that progressively restores manipulated images back to their pristine state in the VAE latent space. By operating independently of the underlying embedding scheme, the proposed method thereby establishes a repair framework for GAN-based deepfakes, while requiring no additional training and remaining compatible with any watermark encoder-decoder. Experimental results show that, even after three consecutive attacks, watermark extraction accuracy remains above 90% when recent SOTA models are augmented with our approach, demonstrating strong robustness within this setting.

Index Terms—deepfake, Media Forensics, Image Recovery

I. INTRODUCTION

Driven by advances in model architectures [1]–[6] and the availability of high-quality datasets [7]–[10], deepfake technology that alters facial characteristics or identity has rapidly evolved and empowered a wide range of applications. In this paradigm, the original image and target label are jointly encoded into a conditional representation, enabling generation tasks such as face swapping and expression manipulation. While deepfake models serve legitimate purposes including identity anonymization [11], [12] and entertainment [13], they can also be exploited by malicious adversaries to spread online rumors or disseminate explicit content without the consent of the original image or video owners. As a result, there is an increasingly urgent need for effective standardization and regulation.

As a classic and widely adopted method, watermarking technique have been extensively used in deepfake forensics [14] due to their invisibility and robustness. Existing watermarking methods support a range of forensic applications, including tampering localization [15], image ownership tracing [16] and direct deepfake detection [17], [18].

However, deepfake models reconstruct images through cumulative manipulations, which can severely damage embedded watermark and hinder reliable extraction during forensic analysis. Generally speaking, the rapid advancement of image generation and manipulation often outpaces the robustness of existing watermarking techniques, leading to an inherent trade-off between visual quality and robustness: strong watermark compromise invisibility, while weak ones are easily destroyed.

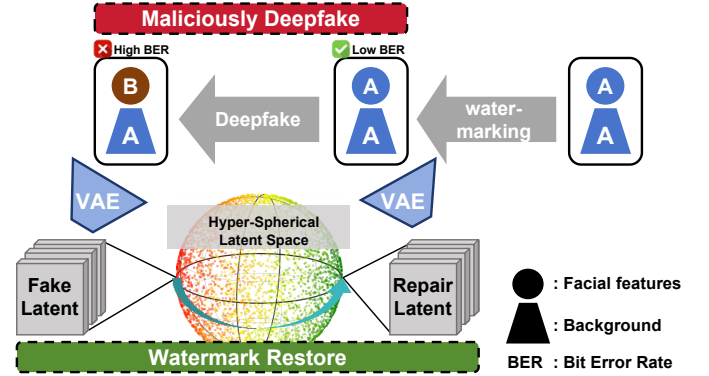


Fig. 1: Our method is to boost watermark ACC via restore fake image on VAE latent space.

This dilemma motivates an alternative perspective: keeping watermark unobtrusive by default and selectively enhancing them only when forensic analysis is required.

To build this framework, we propose ReMark (Fig. 1), a plug-and-play module that restores forged images to an unaltered state through collaborative repair of the image and watermark prior to watermark extraction, thereby improving extraction accuracy. The transformation between fake and original images is governed by complex neural mappings and is difficult to describe explicitly. By leveraging VAEs [19]–[21] to embed both images into a latent space, this complex transformation can be reformulated as a modelable spherical path, along which latents with similar watermark extraction accuracy tend to cluster. We then view restoration as an integration process from the fake latent to the original latent, and train a U-Net [22], [23] to drive this process step by step. Once deployed, ReMark significantly improves watermark extraction accuracy **without modifying any existing watermarking framework or requiring post-training**.

Our main contributions are summarized as follows:

- We find a uniform approach to convert complex path on image domain to modelable VAE latent space.
- We introduce ReMark, a plug-and-play watermark restoration framework that preserves watermark invisibility by default and enables on-demand enhancement for reliable forensic extraction under deepfake attacks.
- Experiments have demonstrated that ReMark can ensure an accuracy of over 90% for the images embedded through existing watermarking models even after 3 times severe deepfakes based on StarGAN [1].

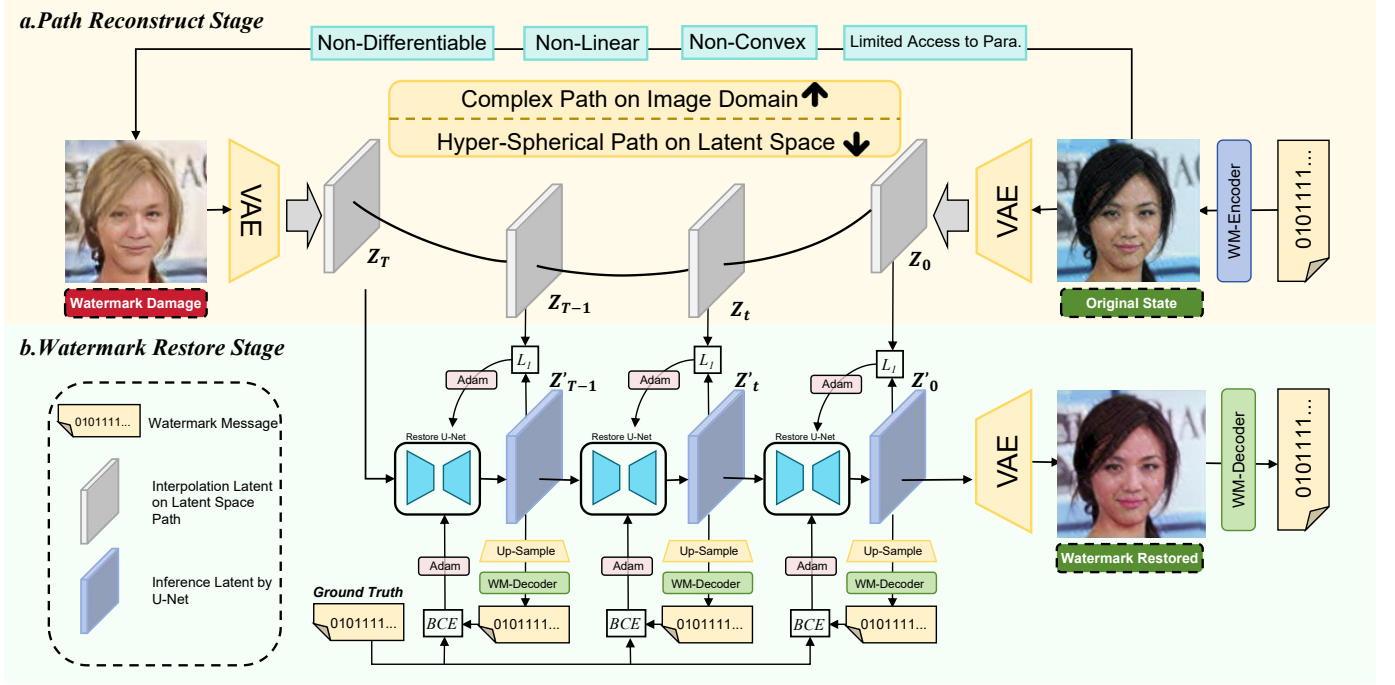


Fig. 2: **Overview of our work:** ReMark is a watermark restore method after the watermark is damaged by malicious deepfake. **(Above)**To complete the repair task, we transform the damage paths that are difficult to describe in mathematical language into controllable paths through latent space. **(Below)**Then a U-Net is trained by opulent path and intermediate point data from stage a, driving the fake latent moves towards original latent and ensuing watermarks restored.

II. RELATED WORK

A. Deepfake

deepfake technology relies on a series of generative deep neural network models; however, unlike conventional Noise-to-Image generation tasks, its core process typically falls under image-to-image conditional generation.

In VAE-based models, such as [4], [6], the input image is first encoded into a latent representation, after which the target label is incorporated as a conditional embedding during decoding to guide the direction of forgery generation. GAN-based models, including [1], [2], typically encode the source image and target label jointly into a conditional vector and synthesize the target image through a complex up-sampling process. In contrast, diffusion-based models [3], [5] integrate the image and target label into a unified condition that guides the reverse denoising process.

All three mainstream generative paradigms share a common mechanism: the image must be down-sampled to form a compact latent representation. However, while this compression facilitates global semantic abstraction, it inevitably leads to the loss of fine-grained details, thereby posing potential threats to the integrity of embedded watermark information.

B. Deepfake Forensics

Evidently, when deepfake technology is exploited for malicious purposes, it can give rise to issues including online misinformation, fraud, and identity abuse. Several approaches have been proposed for the regulation and defense against

harmful deepfake attacks. For instance, adversarial training-based methods [24] introduce perturbations at the encoder level to degrade deepfake outputs; however, such strategies may also interfere with benign deepfake applications. Similarly, forgery detection methods [25] are primarily designed to determine whether an image has been manipulated, but typically provide limited forensic information and cannot reveal critical details such as tampered regions, image ownership, or the identity of the forgery model. To address these limitations, proactive forensic techniques based on invisible watermarking have emerged as a promising solution for the regulation and traceability of deepfake content [26], [27].

Forgery forensics commonly relies on image watermarking techniques. For instance, methods such as [15] compare extracted watermark before and after manipulation to generate masks that indicate tampered regions. Building on this idea, [17] enables both forgery detection and identity tracing by comparing embedded implicit facial landmarks with explicit landmarks visible in marked image. In addition, approaches including [18] embed invisible model-related information within images to support provenance tracking.

However, when the embedded watermark is degraded or destroyed during the processes of dissemination or re-editing, its forensic functionality is consequently lost, thus undermining the reliability of watermark-based forensic systems.

C. Invisible Deep Watermarks

Invisible watermarking techniques, owing to their imperceptibility, have become a crucial component in deepfake forensics. A typical watermarking framework consists of three primary modules—an encoder, a noise layer, and a decoder. Existing frameworks have demonstrated strong robustness against common image edits encountered during media transmission, such as JPEG compression and cropping [28], [29].

However, with the introduction of deepfake, conventional noise layers have become insufficient for protecting watermark. Unlike pixel-level distortion such as gaussian noise or JPEG compression—which do not alter image structures—deepfake models perform large-scale reconstructions that often lead to information loss in fine details, ultimately causing extraction failure.

To address this issue, recent studies have focused on improving watermark robustness and extraction accuracy under deepfake attacks. For example, [30] designs a lightweight additive deepfake noise layer that preserves watermark accuracy while avoiding visual artifacts caused by heavy adversarial training. In addition, [17] enhances watermark resilience by more tightly coupling watermark information with facial landmarks. Furthermore, [15] proposes a more generalizable strategy that embeds only representative up-sampling and down-sampling components of deepfake architectures into the noise layer, rather than incorporating the entire model during training.

Nonetheless, most existing methods emphasize stronger embedding schemes while paying limited attention to how deepfake operations degrade watermark integrity. As deepfake models become increasingly complex, even reinforced embeddings may fail, rendering such defenses largely passive and reactive.

III. METHODOLOGY

A. Proposed Framework

As discussed in the previous section, different deepfake models exhibit similar watermark information loss induced by their reconstruction modules. We posit that watermark degradation caused by deepfake transformations is better understood as a process of concealment rather than erasure, which forms the key theoretical basis of our approach.

This study aims to develop a plug-and-play module that is broadly compatible with existing watermarking frameworks. When watermarks are degraded by deepfake manipulation, the proposed module can effectively restore them to a high level of extraction accuracy. Specifically, the framework consists of two sequential stages:

Firstly, path reestablishment stage: A Variational Autoencoder (VAE) combined with spherical interpolation is used to construct a trajectory from the watermark-degraded latent to the original latent representation. This reconstructed trajectory provides a structural guide for the subsequent watermark restoration stage.

Secondly, watermark restoration stage: Guided by the trajectory established in first stage, a U-Net is trained to

Algorithm 1: Restore Path Reconstruction (VAE)

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1: Input: marked image set  $\mathcal{D} = \{x_i\}$ , deepfake  $\text{DF}(\cdot)$ , repeats  $n \in [0, \bar{n}]$ 
2: Output: VAE encoder/decoder  $(E_\phi, D_\theta)$ 
3: for  $x \in \mathcal{D}$  do
4:   sample  $n \sim \mathcal{U}\{0, \bar{n}\}$ ;  $\tilde{x} \leftarrow \text{DF}^{(n)}(x)$ 
5:   add  $(x, \tilde{x})$  to train set  $\mathcal{T}$ 
6: end for
7: train  $(E_\phi, D_\theta)$  on  $\mathcal{T}$  with loss  $\lambda_1 \mathcal{L}_{\text{L1}} + \lambda_2 \mathcal{L}_{\text{LPIPS}} + \lambda_3 \mathcal{L}_{\text{BCE}} + \beta \mathcal{L}_{\text{KL}}$  until convergence

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iteratively refine the degraded latent toward its original state. The number of inference iterations is adapted to the degradation level, with more severely corrupted latent representation requiring additional refinement steps.

B. Path Reestablishment Stage (Alg. 1)

In the image domain, a deepfake transformation can be expressed as $\text{fake_image} = \text{deepfake}(\text{original_image})$, where $\text{deepfake}(\cdot)$ denotes a model with highly complex nonlinear mappings. In practice, as illustrated by the characteristics of deepfake models shown at the top of Fig. 2, it is difficult to explicitly describe this mapping or invert a forged image back to its original form. To address this challenge, we employ a Variational Autoencoder (VAE) to reformulate the transformation as a modelable path in the latent space.

A VAE with an isotropic Gaussian prior, once trained to convergence, yields latent representations whose aggregated posterior closely approximates $\mathcal{N}(0, I)$. At this stage, each latent dimension behaves approximately as an independent one-dimensional standard normal variable, implying that latent codes are statistically distributed in a spherical manner (see Appendix for details). As a result, most latent samples concentrate near the surface of a high-dimensional hypersphere with an expected radius of approximately $\sqrt{d}$, where d denotes the dimensionality of the latent space.

1) *Loss function:* Building upon the standard VAE training framework, which includes an $\mathcal{L}_{\text{L1}}$ reconstruction loss and a KL divergence term, we further incorporate LPIPS and BCE loss terms to encourage the preservation of watermark information during latent-space encoding. The resulting loss function defines the objective of the first stage in our framework:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{L1}} + \lambda_2 \mathcal{L}_{\text{LPIPS}} + \lambda_3 \mathcal{L}_{\text{BCE}} + \beta \mathcal{L}_{\text{KL}}, \quad (1)$$

where each term serves a distinct role in the optimization process:

- **Reconstruction Loss** ($\mathcal{L}_{\text{L1}}$ and $\mathcal{L}_{\text{LPIPS}}$): Combines pixel-level reconstruction fidelity and perceptual similarity, guiding the decoder to accurately reconstruct the input image while preserving high-level semantic consistency and visual realism.
- **Watermark Loss** ($\mathcal{L}_{\text{BCE}}$): Computed between the extracted watermark W_{fake} from the forged image and the

Algorithm 2: Watermark Repair (U-Net)

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1: Input:  $(E_\phi, D_\theta)$ , U-Net  $G_\psi$ ,  $\text{DF}(\cdot)$ , schedule  $m = x(n)$ ;  
    $n \in [1, \bar{n}]$   
2: Output: restored latent  $\hat{z}$ , recovered watermark  $\hat{W}$   
3: for  $x \in \mathcal{D}$  do  
4:   sample  $n$ ;  $x^{(n)} \leftarrow \text{DF}^{(n)}(x)$   
5:    $z_{\text{src}} \leftarrow E_\phi(x^{(n)})$ ,  $z_{\text{tgt}} \leftarrow E_\phi(x)$   
6:    $\{z_0, \dots, z_m\} \leftarrow \text{SLERP}(z_{\text{src}}, z_{\text{tgt}}, m)$   
7:   for  $k = 0$  to  $m - 1$  do  
8:      $\tilde{z}_{k+1} \leftarrow G_\psi(z_k)$   
9:     update  $G_\psi$  by minimizing  
        $\lambda_1 \|\tilde{z}_{k+1} - z_{k+1}\|_1 + \lambda_2 \text{BCE}(W_{\text{gt}}, \text{Ext}(D_\theta(\tilde{z}_{k+1})))$   
10:  end for  
11:   $\hat{z} \leftarrow \tilde{z}_m$ ;  
    $\hat{x} \leftarrow D_\theta(\hat{z})$ ;  
    $\hat{W} \leftarrow \text{Ext}(\hat{x})$   
12: end for
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ground truth watermark W_{gt} using binary cross-entropy, encouraging reliable watermark recovery.

- **Regularization Term ($\mathcal{L}_{\text{KL}}$):** Enforces the posterior distribution $q_\phi(z|x)$ to remain close to the isotropic gaussian prior $\mathcal{N}(0, I)$, thereby promoting a well-structured and continuous latent space.

After training, latent representations with similar watermark extraction accuracy tend to cluster in the latent space. We connect sample pairs from two such clusters along the hyperspherical surface and generate intermediate latent points using spherical linear interpolation (SLERP), defined as:

$$z(t) = \frac{\sin((1-t)\theta)}{\sin \theta} z_1 + \frac{\sin(t\theta)}{\sin \theta} z_2, \quad t \in [0, 1], \quad (2)$$

where θ denotes the angle between latent vectors z_1 and z_2 , and increasing t indicates progressive loss of watermark information..

It is worth noting that, despite focusing on watermark recovery accuracy, we keep an image semantic reconstruction term in the loss, since most binary watermarks (e.g., identity-traceability watermarks) are tightly coupled with image content during embedding.

C. Watermark Restoration Stage(Alg. 2)

In the first stage, we extract representative latent trajectories that describe how degraded latents map back to their pristine counterparts. In the second stage, guided by these trajectories, we train a U-Net to iteratively refine degraded latents, with the number of inference steps adjusted to the degradation level. This adaptive restoration drives latents toward their original states, enabling reliable watermark recovery with low bit error rate.

To model the process of watermark restoration, we introduce a **time-dependent dynamical system** that describes the recovery of watermark information in the latent space via an ordinary differential equation (ODE). Specifically, the temporal evolution of the latent variable $z(t)$ is defined as:

$$\frac{dz}{dt} = f_\theta(z, t) = -\lambda(t) [z(t) - g_\theta(z(t), t)], \quad (3)$$

where $f_\theta(z, t)$ represents the parameterized restoration model, $\lambda(t)$ denotes a time-varying dynamic restoration rate, and g_θ corresponds to the predictive function that estimates the next-step restoration based on the current state.

By solving above differential equation, the restoration process can be expressed in an **integral form** as:

$$z(0) = z(T) + \int_T^0 f_\theta(z(t), t) dt. \quad (4)$$

In practice, performing continuous integration along the hyperspherical path with infinitely many evaluation points is computationally infeasible. Therefore, we adopt a discrete approximation that more closely reflects realistic computational settings, given by:

$$z_0 \approx z_T + \sum_{k=1}^N \Delta t_k f_\theta(z_{t_k}, t_k), \quad (5)$$

where N denotes the number of discrete interpolation steps.

As the number of interpolation points increases, the discrete formulation naturally converges to the continuous integral in Eq. 4. Leveraging the inherent convexity and continuity of the reconstructed trajectory, our method is able to capture essential restoration dynamics with minimal interpolation steps. This property allows us to achieve high restoration accuracy while significantly reducing computational overhead.

Distinct from $f_\theta(z, t)$ which models the instantaneous latent variation rate (Eq. 3), we define g_θ as a finite-step approximation that directly predicts the subsequent latent state. This discrete formulation effectively mitigates numerical accumulation errors often associated with continuous integration. Given the symmetric input-output structure of this mapping, we employ a U-Net architecture, where the inherent inductive bias provided by skip connections is well-suited for preserving spatial fidelity during the restoration process.

1) *Loss function:* For the U-Net model, we consider two primary objectives: ensuring accurate latent restoration and maintaining reliable watermark recovery. Accordingly, the loss function is defined as:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{L1}} + \lambda_2 \mathcal{L}_{\text{BCE}}. \quad (6)$$

Each term in the objective contributes to a distinct aspect of model optimization:

- **Latent Reconstruction Loss ($\mathcal{L}_{\text{L1}}$):** Measures the absolute difference between the input latent z_{input} and the output latent z_{output} , encouraging precise spatial restoration and directional consistency within the latent space.
- **Watermark Consistency Loss ($\mathcal{L}_{\text{BCE}}$):** Computes the binary cross-entropy between the recovered watermark W_{rec} and the ground-truth watermark W_{gt} , enforcing reliable watermark recovery and improved extraction accuracy after restoration.

deepfake model	Times	HiDDeN(ECCV'19) [29]			MBRS(MM'21) [28]			SepMark(MM'23) [30]			FIN(AAAT'23) [16]			LampMark(MM'24) [17]		
		ACC	ACC*	Δ	ACC	ACC*	Δ	ACC	ACC*	Δ	ACC	ACC*	Δ	ACC	ACC*	Δ
StarGAN	0	92.97	—	—	100.00	—	—	100.00	—	—	100.00	—	—	99.71	—	—
	1	57.12	92.56	35.44	91.21	99.65	8.44	96.69	98.12	1.43	78.71	99.65	20.94	98.05	98.95	0.90
	2	54.19	92.38	38.19	77.34	99.52	22.18	93.01	95.71	2.70	67.19	99.52	32.33	91.31	98.48	7.17
	3	52.53	92.26	39.73	65.43	98.90	33.48	81.41	90.04	8.63	64.75	98.90	34.15	83.64	97.97	14.33

TABLE I: Watermark extraction accuracy (%) for different encoder-decoder methods. **ACC*** denotes the extraction accuracy after enhancement by the proposed boosting module. Times = 0 indicates that no manipulation is applied, and the watermark is extracted directly after embedding.

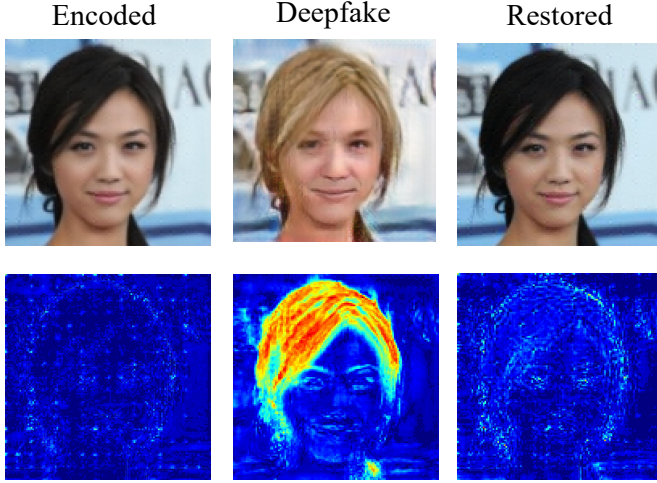


Fig. 3: Residual visualization of encoded, deepfake, and restored images with **MBRS** watermark embedding. These cross-shaped highlights indicate the watermark signal.

IV. EXPERIMENTS

A. Experiments Setup

We train the Path Model (VAE) and the Restore Model (U-Net) on the CelebA dataset [7], using 10,000 images for training and 2,000 images for validation, all resized to $128 \times 128 \times 3$. Both models are optimized with the Adam optimizer, with an initial learning rate of 10^{-5} , a batch size of 16, and a fixed learning-rate schedule. For spherical linear interpolation (SLERP) in Eq. 2, we employ 10 intermediate latent points for each degradation level. The KL-VAE adopts a down-sampling factor of 2^4 , resulting in a latent resolution of $16 \times 16 \times 4$. All experiments are conducted on a single NVIDIA RTX 4090 GPU.

B. Comparison with Forensics Methods

To evaluate the gain brought by our restoration module, we compare it with several representative invisible watermarking baselines: HiDDeN [29] and MBRS [28] (DNN-based methods for common edits), FIN [16] (a flow-based model for diverse transformations), and SepMark [30] and LampMark [17] (methods tailored for deepfake-oriented degradations). We use watermark extraction accuracy (ACC) as the primary metric. Since our goal is message recovery after restoration rather

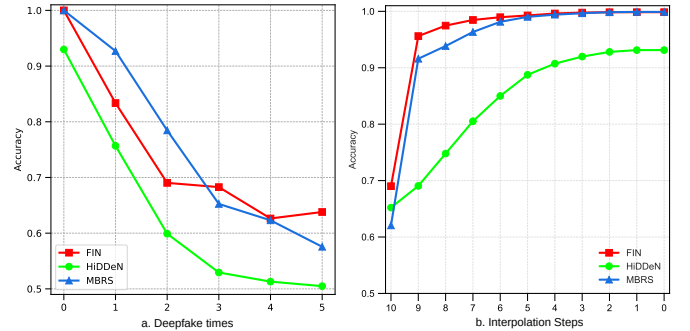


Fig. 4: Accuracy trends for deepfake times and interpolation path.

than perceptual quality of restored images, PSNR/SNR/SSIM are not treated as core metrics.

To simulate cumulative manipulations, we adopt a pre-trained StarGAN [1] as the deepfake operator, as GAN-based attribute translation has been widely observed to induce strong distribution shifts and resampling artifacts that severely degrade embedded watermark signals, providing a practical worst-case stress test for Internet-scale deepfake propagation.

C. Effect of Enhancing

Most baseline watermarking methods embed a 64-bit binary message (except LampMark); under this setting, an extraction accuracy close to 50% indicates near-complete corruption. As shown in Table. I, before applying ReMark, HiDDeN, MBRS, and FIN (without dedicated deepfake-robustness training) drop below 65% accuracy after three consecutive deepfake transformations, while the remaining baselines fall below 85%. Such performance is insufficient for practical deployment, especially when content may be manipulated multiple times during Internet dissemination.

With ReMark reconstruction, extraction accuracy rises to over 95% for almost all baselines, except HiDDeN due to inherent architectural limitations. Fig. 3 shows that deepfake outputs (**middle**) heavily smooth watermark-related structures, obscuring the low-frequency cues needed for decoding, whereas the reconstructed results (**right**) restore the cross-shaped highlight patterns in residual maps, providing enough signal for reliable extraction and substantially lower BER.

D. ACC trends in Latent Space

As shown in Fig. 4a, extraction accuracy consistently declines under consecutive deepfake manipulations, motivating our SLERP-based latent path and restoration strategy. Fig. 4b shows that ACC evolves smoothly along the path, increases monotonically, and converges to 100%, supporting the feasibility of path integration driven by a U-Net.

The number of SLERP intermediate points is a hyperparameter: more points yield a smoother, easier-to-fit path but incur higher computation. In practice, using 5 or 15 points shows no significant difference, likely due to the convexity and monotonicity of SLERP. Balancing accuracy and cost, we adopt 10 points.

E. Future Work

This work primarily evaluates the proposed framework under GAN-based deepfake models. We also conduct preliminary experiments on diffusion-based models using the MBRS embedding method, where watermark extraction accuracy can be improved by approximately 10%–15%. However, the absolute performance remains limited, indicating that the current restoration formulation is not yet fully adapted to diffusion-based generation mechanisms.

An important direction for future work is to extend the proposed framework to diffusion-based deepfakes by designing restoration dynamics tailored to diffusion processes, with the goal of achieving more general and robust watermark repair across different deepfake paradigms.

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